

Pseudo-Spherical Contrastive Divergence

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What is Distribution Learning?

Supervised learning vs Generative modeling

Supervised: learn $p(y|x)$ from pairs (input, target). Generative: learn $p(x)$ directly — no labels, just examples.

The challenge

How do we compare two distributions? Standard metric: KL divergence.
But this comes with hidden costs...

Energy-Based Models and the Partition Function

EBM formula

$$p_{\theta}(x) = \frac{\exp(-E_{\theta}(x))}{Z_{\theta}}, \quad Z_{\theta} = \int \exp(-E_{\theta}(x)) dx$$

Low energy $\rightarrow$ high probability. High energy $\rightarrow$ low probability.

The core problem

Z_{θ} is an integral over all possible data (dimension ~ 3000 for 32×32 image). **Impossible to compute** (intractable).

Consequence

Standard MLE requires $\nabla_{\theta} \log Z_{\theta}$, which involves Z . We cannot compute it directly.

Standard Solution: Contrastive Divergence

The MLE gradient splits into two parts

$$\nabla_{\theta} \mathcal{L}_{MLE} = \mathbb{E}_{x \sim p_{\text{data}}} [\nabla_{\theta} E(x)] - \mathbb{E}_{x \sim p_{\theta}} [\nabla_{\theta} E(x)]$$

“Pull down” energy of real data, “push up” energy of model samples.

Problem: sampling from p_{θ}

Use MCMC (Langevin Dynamics): iteratively update $x^{(t+1)} = x^{(t)} - \frac{\epsilon}{2} \nabla E(x^{(t)}) + \sqrt{\epsilon} z^{(t)}$.

Issues

MCMC is slow. KL-divergence is zero-forcing: if $p(x) > 0$ but $q(x) \approx 0$, then $\text{KL} = \infty$. Model must fit outliers. Sensitive to data contamination.

Scoring Rules: Old Idea, New Application

Definition

Scoring rule $S(x, q)$ assigns a score to prediction q given outcome x .

Strictly proper: $\mathbb{E}_{x \sim p}[S(x, p)] > \mathbb{E}_{x \sim p}[S(x, q)]$ unless $q = p$.

Intuition: tells the truth about the distribution.

Example: log score

$S(x, q) = \log q(x)$. Expected score is $-\mathbb{E}[-\log q(x)]$, which is related to KL divergence.

The insight

Are there other strictly proper scoring rules that don't require computing Z ?

Homogeneity: The Magic Property

Homogeneous scoring rule

$$S(x, \lambda \cdot q) = S(x, q) \quad \text{for all } \lambda > 0$$

Scaling the distribution doesn't change the score.

Why this solves everything

Unnormalized density: $\tilde{q}(x) = \exp(-E(x))$. Normalized: $q(x) = \tilde{q}(x)/Z$.

By homogeneity:

$$S(x, q(x)) = S\left(x, \frac{\tilde{q}(x)}{Z}\right) = S(x, \tilde{q}(x))$$

The partition function Z cancels! We only need the energy.

Pseudo-Spherical Scoring Rule

Definition, parameterized by $\gamma > 0$

$$S(x, q) = \frac{q(x)^\gamma}{\|q\|_{\gamma+1}^\gamma}, \quad \|q\|_{\gamma+1} = \left(\int q(y)^{\gamma+1} dy \right)^{1/(\gamma+1)}$$

For numerical stability, use log version:

$$L_\gamma(p, q) = \frac{1}{\gamma} \log \mathbb{E}_p[q(x)^\gamma] - \log \|q\|_{\gamma+1}$$

Properties

Strictly proper, homogeneous, parameterized by γ .

- ▶ Large γ : focus on quality (mode-seeking).
- ▶ Small/negative γ : focus on diversity (mode-covering).

Algorithm: PS-CD

The objective

$$\min_{\theta} L_{\gamma}(\theta; p) = -\frac{1}{\gamma} \log \mathbb{E}_p[\exp(-\gamma E_{\theta}(x))] - \log \|q_{\theta}\|_{\gamma+1}$$

The trick: importance sampling

Sample only from q_{θ} using Langevin Dynamics. Use weights $w(x) = \exp(-\gamma E_{\theta}(x))$ to estimate expectations under auxiliary distribution.

Low energy $\rightarrow$ weight ≈ 1 (important). High energy $\rightarrow$ weight ≈ 0 (less important).

Gradient update

$$\nabla_{\theta} L_{\gamma}^N = -\nabla_{\theta} \frac{1}{\gamma} \log \left(\frac{1}{N} \sum_i \exp(-\gamma E(x_i^+)) \right) - \frac{\sum_i w_i \nabla_{\theta} E(x_i^-)}{\sum_i w_i}$$

PS-CD: From Theory to Practice

Algorithm steps

1. Draw batch of real data $\{x_1^+, \dots, x_N^+\} \sim p_{\text{data}}$.
2. Generate fake samples $\{x_1^-, \dots, x_N^-\}$ from q_θ using Langevin Dynamics.
3. Compute importance weights: $w_i = \exp(-\gamma E_\theta(x_i^-))$.
4. Update: standard SGD with the weighted gradient above.

Comparison to classical CD

Same MCMC sampling. Different loss (γ -score, not KL). Weighted gradients.

Connection to MLE

Key fact: when $\gamma \rightarrow 0$, PS-CD recovers classical CD. Classical CD is a special case!

Robustness to Data Contamination

Experiment: Gaussian data + noise

True: $\mathcal{N}(-1, 0.5)$. Contamination: $\mathcal{N}(2, 0.05)$ with ratio $c \in [0.01, 0.3]$.

Metric: $D_{KL}(p_{\text{true}} \| q_{\theta})$ after training.

Contamination	CD	PS-CD ($\gamma = 0.5$)	PS-CD ($\gamma = 1.0$)	PS-CD ($\gamma = 2.0$)
0.01	0.0067	$1e^{-5}$	$1e^{-7}$	$1e^{-6}$
0.05	0.0851	0.00027	$1.6e^{-6}$	0.00011
0.1	0.1979	0.00173	$1.86e^{-6}$	0.00012
0.2	0.3869	0.1858	$6.4e^{-6}$	0.00017
0.3	0.5438	0.5429	0.3118	0.00029

Conclusion

PS-CD with $\gamma = 1.0$ survives 20% contamination. CD fails catastrophically.

Image Generation: CIFAR-10 Results

Task: generate 32×32 images

Metric: FID (Fréchet Inception Distance). Lower is better.

Method	FID (CIFAR-10)
CD (classical, $\gamma = 0$)	37.90
f-EBM (minimax)	30.86
PS-CD ($\gamma = -0.5$)	27.95
PS-CD ($\gamma = 1.0$)	29.78

Key insight

PS-CD achieves better FID than f-EBM **without minimax training**.

Different γ values: quality-diversity trade-off.

The γ Parameter: Quality-Diversity Trade-off

1D synthetic example: fit mixture of Gaussians with single Gaussian

- ▶ $\gamma = -0.5$: Covers both modes poorly (mode-covering).
- ▶ $\gamma = 0$ (KL): One accurate mode (classical MLE).
- ▶ $\gamma = 1.0$ (spherical): Very tight fit to one mode (mode-seeking).

Practical guidance

- ▶ **Diversity needed** (text generation): use $\gamma < 0$ or $\gamma \approx -0.5$.
- ▶ **Quality needed** (density estimation): use $\gamma > 0$ or $\gamma = 1.0$.
- ▶ **Balanced**: $\gamma \in (0, 1)$.

Out-of-Distribution Detection

Task

Train on CIFAR-10. Test: can we distinguish in-distribution from OOD (SVHN, noise, etc.)? Score: $s(x) = \max_y [-E(x, y)]$ (negative energy).
Metric: AUC (higher is better).

Results

PS-CD consistently outperforms CD across OOD datasets (SVHN, Textures, Noise).

Why it matters

Beyond generation, PS-CD models are better at anomaly detection, outlier rejection, and other density-related tasks. The learned energy function has better in-distribution / OOD separation.

Why PS-CD Works: Key Advantages

Homogeneous scoring rules

Eliminate Z from all equations. Only need unnormalized energy.

Parameter γ

Flexible control: quality-diversity trade-off without retraining.

Importance weighting

Model self-determines which negative samples matter. Focuses learning on important regions.

No minimax

Single network, single objective, stable training (unlike GAN or f-EBM).

Robustness

Not zero-forcing (unlike KL). Handles data contamination and outliers better.

Conclusions and Main Contributions

The Problem

EBMs powerful, but hard to train due to intractable partition function Z .
Classical CD slow, unstable, sensitive to contamination.

The Solution

Use strictly proper homogeneous scoring rules (pseudo-spherical, parameterized by γ). Eliminates Z . Enables flexible quality-diversity trade-offs.

The Algorithm

PS-CD: MCMC + importance weighting. Same cost as classical CD, benefits of f-EBM without minimax instability.

Validation

Robust to contamination. Better image generation (lower FID). Better OOD detection. Different γ values for different tasks.

Theory

Authors prove convergence guarantees and sample complexity bounds. Classical CD is the special case $\gamma = 0$. PS-CD is a proper generalization with solid mathematical foundations.